



Identification of the viscoelastic response and nonlinear damping of a rubber plate in nonlinear vibration regime

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ABSTRACT

Three different dissipation models have been used to identify the increase of damping with the vibration amplitude for a rubber rectangular plate. For this purpose, a square rubber plate made of silicone with fixed edges has been tested and its linear and nonlinear responses have been measured by laser Doppler vibrometers. First, a reduced-order model, using energy based approach and global discretization, has been constructed, taking into account geometric imperfections; the linear viscous damping at each excitation level in the nonlinear regime has been identified from the experimental data. This numerical model with linear viscous damping has been widely validated and constitutes the basis for comparison with subsequent damping identifications. Then, three different single degree of freedom models have been fitted to the same experimental data; each model has a different damping description. Specifically, the models are based on a modified Duffing oscillators with linear, quadratic and cubic stiffness and: (i) a linear viscous damping; (ii) a nonlinear viscoelastic dissipation described by the loss factor; (iii) a standard linear solid viscoelastic model with nonlinear springs. The dissipation identified by the different models is discussed and confirms the major nonlinear nature of damping as a function of the vibration amplitude.

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1. Introduction

Understanding the structural damping is of critical importance for effective design. The importance increases manifolds when the structure experiences large amplitude vibrations, as its nonlinear dynamics is even more influenced on the amount of damping than in the linear (i.e. small amplitude) vibration regime [1]. Experimental data is mandatory to identify the damping present in a structure, both in linear and nonlinear regime. However, damping cannot be directly measured, so a dissipation model must be introduced in order to identify its value. Dissipation depends, among others, on the geometry, material, surrounding fluid, boundary conditions and vibration mode shape.

There are well established tools available to extract the viscous damping ratio in the linear vibration regime. Experimental modal analysis is one among them and has become the industrial standard decades ago. However, thin walled structures such as plates, panels and shells experience large amplitude vibrations, i.e. vibration amplitude of the order of the thickness giving rise to geometrical nonlinearity, during their normal operating conditions and experimental modal analysis or any

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other tools based on the linear vibration assumption cannot be used to extract the damping of them. Hence there is a clear need for developing tools to extract damping in the nonlinear vibration regime for thin walled structures.

Recent experimental studies show that the damping present in a structure increases nonlinearly as the vibration amplitude increases. This phenomenon is well documented in cantilever beams [2], plates [3,4], panels [3] and shells [5] considering viscous damping model for representing the dissipation energy.

In order to predict the nonlinear dynamics of a structure using a reasonable amount of degrees of freedom, with intrinsic numerical advantages in reliability and computational cost, it is necessary to build a Reduced Order Model (ROM) [6,7]. Often the damping in the ROM is introduced as linear viscous damping, e.g. by using Rayleigh's dissipation function [7]. Developing an accurate ROM for a structure involves numerous complexities. A simple tool to extract the damping of a structure from its experimentally measured nonlinear response, without the need of developing a ROM, would give engineers/scientists clear advantage. To address this need, a tool based on harmonic balance method was developed considering viscous damping model in a modified Duffing nonlinear resonator with quadratic and cubic stiffness terms [8]. This tool successfully extracts the damping ratio present in a structure from its amplitude-response curves at different excitation levels and the results match very well with the damping identified from the sophisticated ROM of the structure. Experimental results show that the increase in damping with the peak vibration amplitude (and the level of harmonic force excitation) is substantially large [8]. Not taking it into account would lead to inefficient design of such structures and very large overprediction of the vibration amplitude. In fact, most of the studies in nonlinear vibration show complicated nonlinear dynamics which is fully destroyed by the increase of the actual damping due to the large amplitude of vibrations.

Structures made of rubber-like or biological materials exhibit substantial viscoelastic behaviour [9]. For these structures, the viscous damping model does not capture accurately the dissipated energy. Viscoelastic damping models such as the Kelvin-Voigt, hysteretic, standard linear solid or Boltzmann models have been used for that purpose [10–13]. A previous study by our research group shows the variation of relaxation time (dissipation parameter of Kelvin-Voigt model viscoelastic model) versus the vibration amplitude of rubber plates [14]. The nonlinear damping introduced by the Kelvin-Voigt model is not sufficient to capture the damping exhibited by the structure during large amplitude vibrations. So, the relaxation time has to be increased as the vibration amplitude increases to match the numerical response with experimental measurements, implying that Kelvin-Voigt model is not sufficient for modelling the nonlinear damping.

Hysteretic damping (where stiffness and damping are represented together by a complex spring) has been used extensively to describe the dissipation present in viscoelastic systems using the loss tangent. For example, the loss tangent of an aluminium plate having viscoelastic core with various fiber orientation was identified experimentally by Berthelot et al. [15]. A novel method to identify the loss tangent in the linear regime was developed by Liu and Ewing [16]. The loss and storage energies were experimentally calculated by dividing the mobility at the driving point by the mobility of the measurement point. The loss tangent was also identified by frequency response functions measured on different points of the geometry for a composite plate [17]. However, the variation of hysteretic damping during large-amplitude vibrations of a structure has not been completely addressed yet, as per authors' knowledge.

It is evident that a nonlinear dissipation model is necessary to predict structural dynamics at large-amplitude vibration [3]. Few researchers have attempted to use phenomenological nonlinear damping for various structures. Zaitsev et al. [18] used the Kelvin-Voigt model for describing the nonlinear damping of micromechanical oscillator and concluded that the model is better than linear viscous damping but not sufficient and further work is required. Gottlieb and Habib [19] used a phenomenological nonlinear damping model to understand the large amplitude, quasi-periodic and chaotic vibrations of a spherical pendulum. Eichler et. al. [20] used a damping model containing a nonlinear term proportional to the square of the vibration amplitude multiplied by the velocity, similar to the one previously introduced in [18], without any derivation; it was applied to estimate the damping in carbon nanotubes and graphene devices. Recently Amabili [21] derived a model of nonlinear damping based on a fractional standard linear solid material after introducing geometric nonlinearity in it. The model was successfully compared to experimental results for vibrations of a plate, a beam and a curved panel in geometrically nonlinear regime.

In the present study, three different dissipation models have been used to identify the increase of damping with the vibration amplitude for a rubber rectangular plate. An overview of the organization of the study with the different models developed is shown in Fig. 1. For this purpose, a square rubber plate made of silicone with fixed edges has been tested and its linear and nonlinear responses have been measured by laser Doppler vibrometers. First, a ROM, using energy based approach and global discretization, has been constructed, taking into account geometric imperfections; the linear viscous damping at each excitation level in the nonlinear regime has been identified from the experimental data. This numerical model with linear viscous damping has been widely validated and constitutes the basis for comparison with subsequent damping identifications. Then, three different single degree of freedom (SDOF) models have been fitted to the same experimental data; each model has a different damping description. Specifically, the models are based on a modified Duffing oscillators with linear, quadratic and cubic stiffness and: (i) a linear viscous damping; (ii) a nonlinear viscoelastic dissipation described by the loss factor; (iii) a standard linear solid viscoelastic model with nonlinear springs. The dissipation identified by the different models is discussed and confirms the major nonlinear nature of damping as a function of the vibration amplitude.

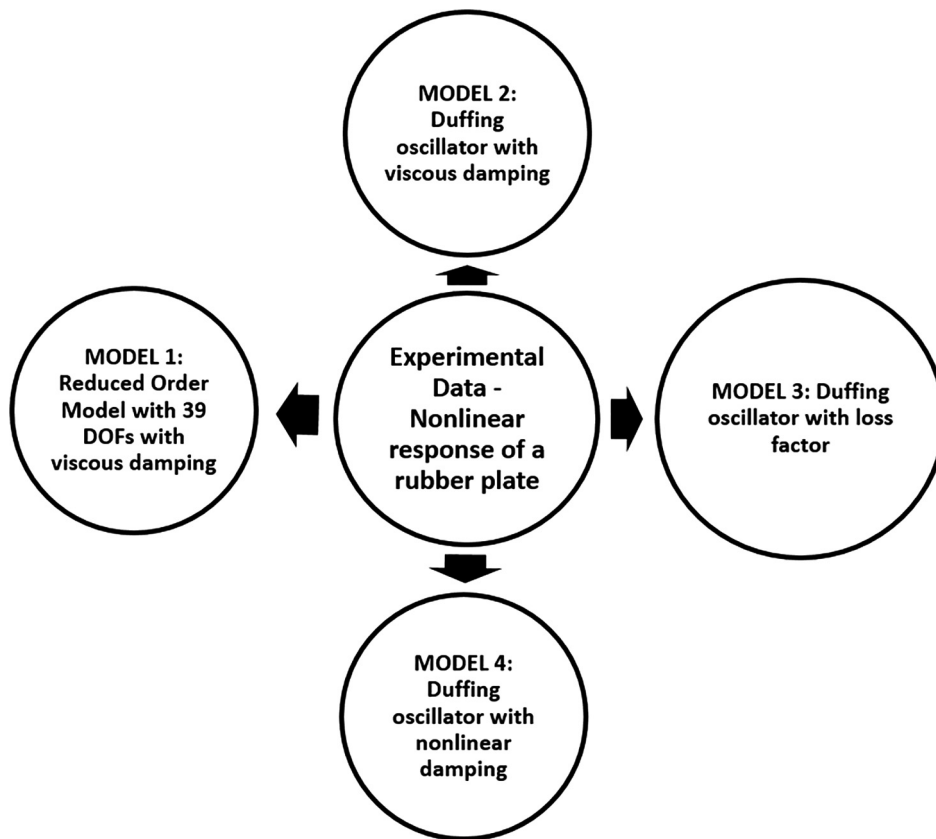


Fig. 1. Different models used to identify the damping from the experimental data.

2. Experimental study on a rubber plate

A thin walled rubber plate made of silicone was chosen for this study. The sheet of rubber was purchased from commercial market with the description of “high-temperature silicone sheet”. The shore hardness is 90A, the density is approximately 1430 kg/m^3 and the thickness is 3.35 mm. Viscoelastic structures such as the plate under study here exhibit frequency dependent complex modulus. To measure the same, a strip of the rubber sheet was cut with the width of 11 mm and length of 94 mm and assembled as fixed–fixed beam without initial pre-tension. The beam was excited laterally (in the direction of the smaller dimension) at middle-length by using an electrodynamic shaker equipped with piezoelectric load cell and stinger. From the beam deflection at various frequencies, the frequency dependent complex modulus is calculated and shown in Fig. 2. The real part of the complex modulus is called the storage modulus (solid line in Fig. 2) and the imaginary part is named the loss modulus (dotted line in Fig. 2). The storage modulus is increasing nonlinearly as the frequency increases. Whereas, loss modulus is increasing linearly with respect to frequency within the frequency range studied. The loss factor of the material, which is given by the ratio between the loss and storage moduli, is shown in Fig. 3. The loss factor increases from about 0.01 at 5 Hz to 0.13 at 100 Hz.

2.1. Rubber plate

A square rubber plate was glued on one side to a heavy metal frame for experiments in large-amplitude vibration regime. The dimensions of the free part of the plate inside the frame were $260 \times 260 \text{ mm}$. The glue ensures fixed boundary conditions in the out-of-plane direction since it is stiffer than the plate in bending. However, the in-plane stiffness of the plate is larger than the combined shear stiffness of the glue layer and the shear stiffness of the silicon rubber. In fact, the plate is glued only on one side, so that in-plane displacements at the edges, assumed to be located on the middle plane of the plate, are allowed by the shear of 1.675 mm of silicon rubber (half the thickness of the plate) and the shear of the glue layer. This ends in approximately obtain free-edge boundary conditions in both the in-plane directions. The glued rubber plate is shown in Fig. 4. Since no in-plane stretch was applied to the rubber plate during gluing, the plate is not perfectly flat and geometric imperfections are present. The geometric imperfections are very important for correctly simulating the nonlinear vibration response of plates and shells. To measure the geometric imperfections of the plate, the FAROARM[®] scanning system was

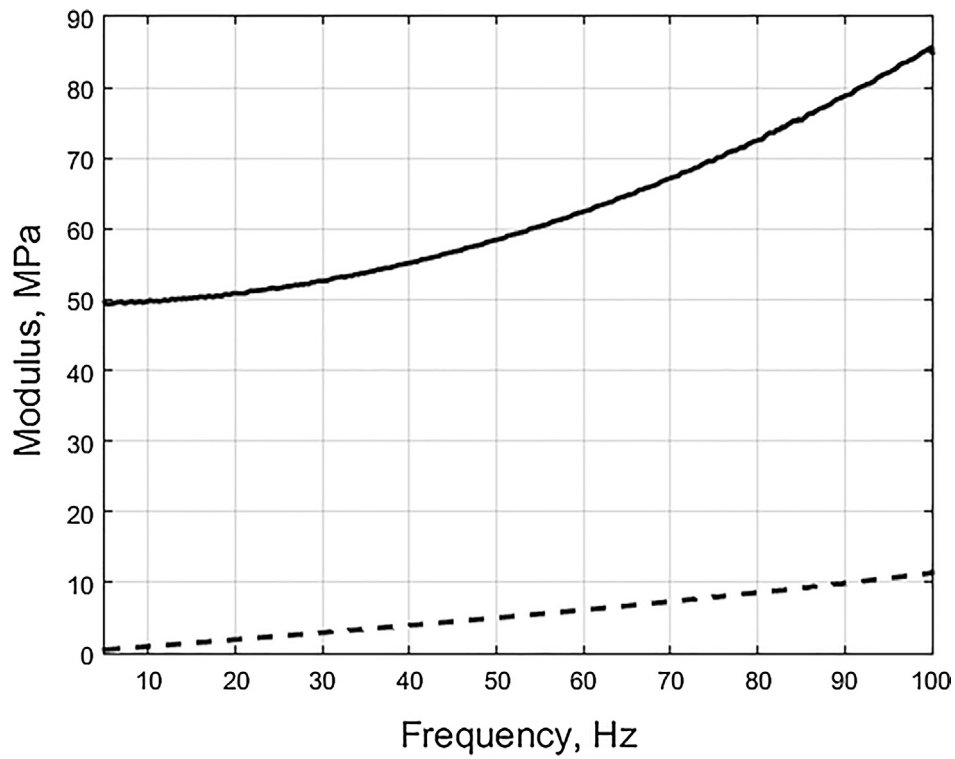


Fig. 2. Identified complex modulus of the silicone rubber. —, Storage modulus; --, loss modulus.

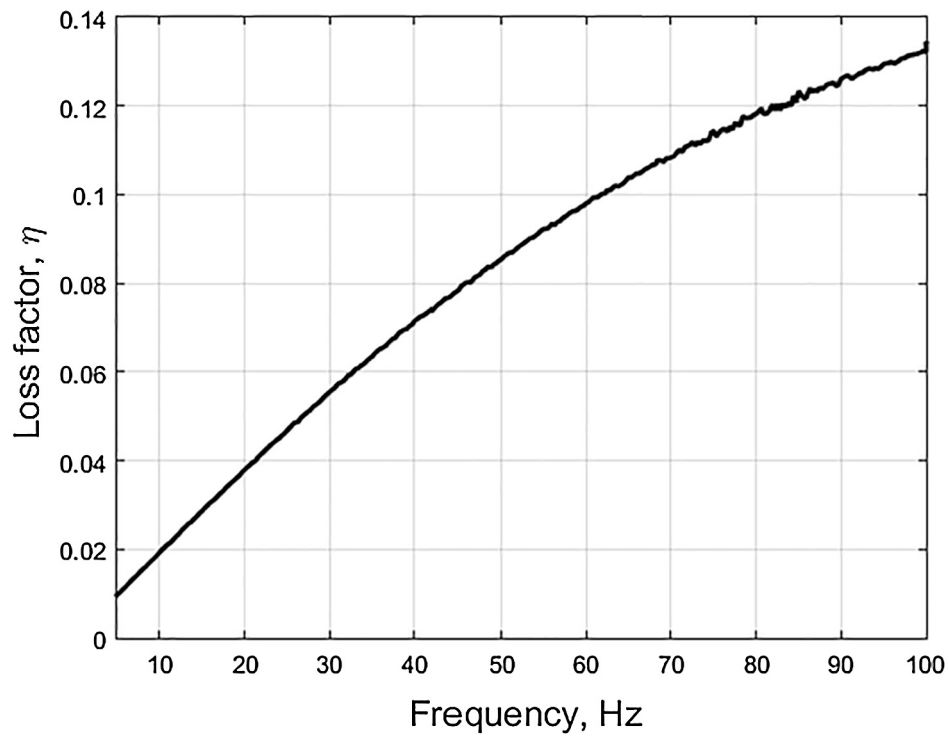


Fig. 3. Identified loss factor of the silicone rubber.

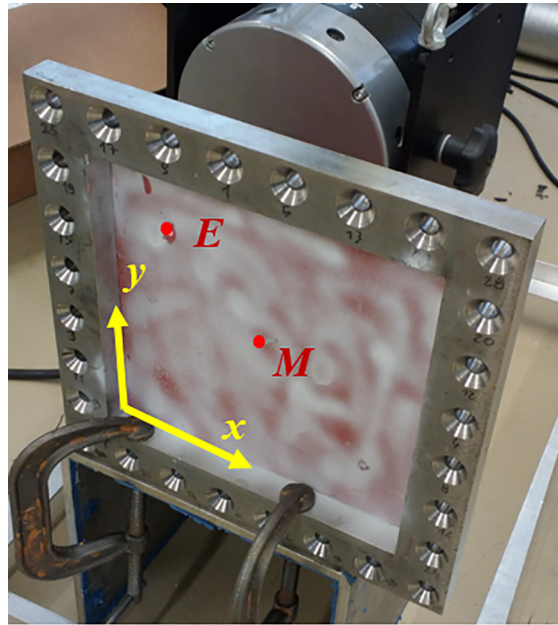


Fig. 4. Silicone rubber plate glued to a metal frame; *E*, Excitation point, *M*, Measurement point. The spray paint is applied to increase reflectivity for the laser measurement.

used. The glued rubber plate was kept on the scanning table and the profile of the rubber plate was scanned. The scanned points were imported and the 3D image of the surface of the plate was constructed as shown in Fig. 5.

2.2. Measurement setup

For dynamic tests, the structure was excited by means of transversal punctual excitation using a thin stinger connected to an electrodynamic exciter (Brüel & Kjær model 4824). The excitation was positioned close to the top left corner, at a distance of 50 mm from the left vertical edge and 60 mm from the top horizontal edge (point *E* in Fig. 4). This excitation position is chosen in such a way that all the low-frequency fundamental modes are excited and there is little interaction between the shaker and the plate during large amplitude vibrations. A very small force transducer (Brüel & Kjær model 8203) glued to the

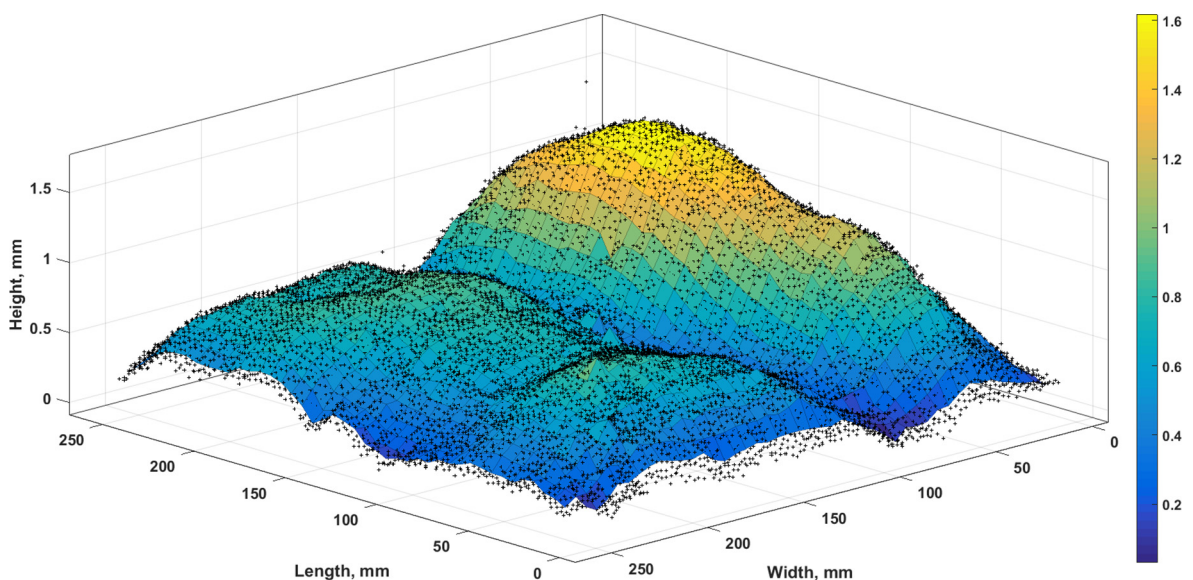
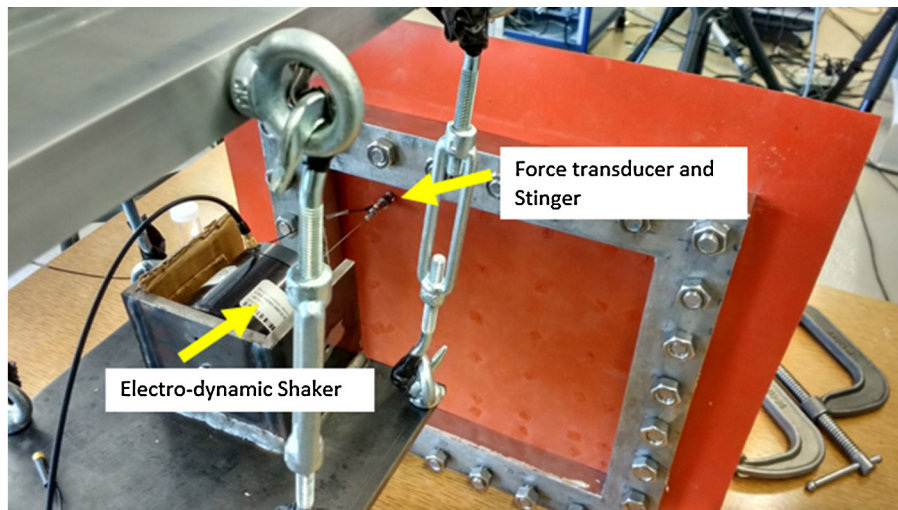


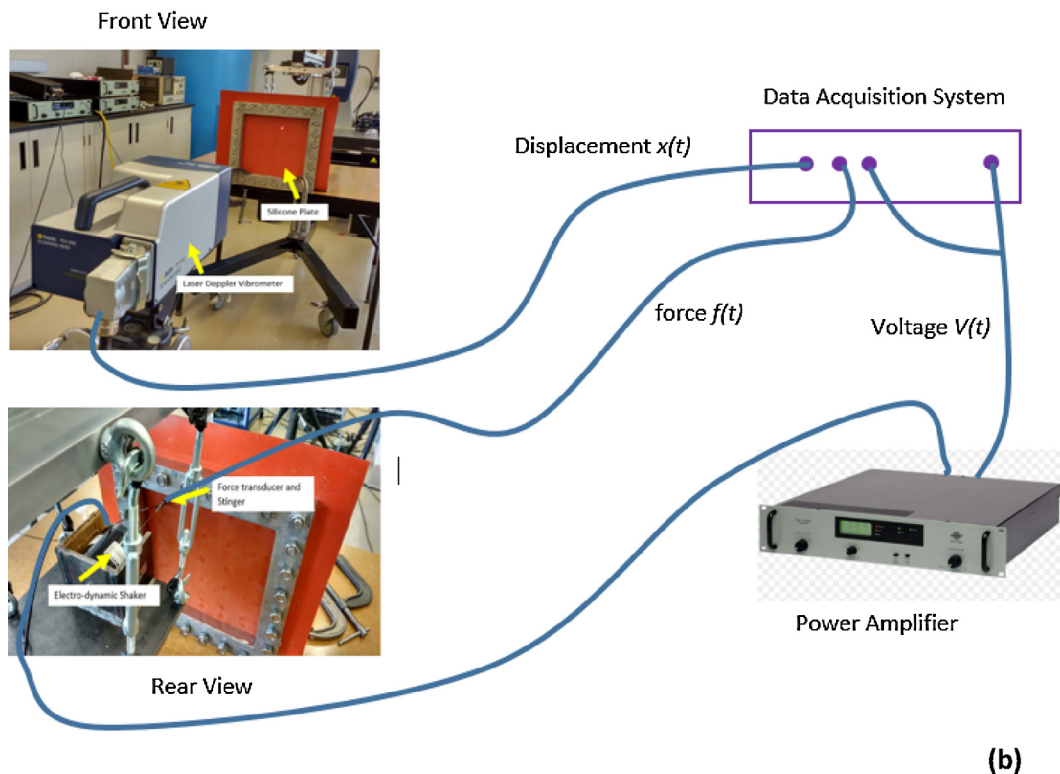
Fig. 5. Measured surface of the glued rubber plate, which gives the geometric imperfections.

plate was used to measure the force applied and it was connected to the electrodynamic exciter through a stinger, as shown in Fig. 6(a).

In the linear experiments, an automatic scanning laser Doppler vibrometer (Polytec PSV 400) was used to measure the vibration. The scanning laser system was employed to measure the Frequency Response Functions (FRFs) at a large number of predefined points covering the whole plate. The plate was excited with pseudo-random signal in the linear study as it eliminates the leakage effect. The measured FRFs were exported to LMS Test.Lab software for modal parameter extraction using the PolyMAX algorithm [22].



(a)



(b)

Fig. 6. Measurement setup for the vibration experiments on the rubber plate. (a) Electro-dynamic exciter, piezoelectric load transducer and stinger; (b) overview of the experimental set-up.

During large-amplitude vibration experiments, a single-point laser vibrometer (Polytec OFV 505) was used. The laser was pointed at the centre of the plate, since the fundamental mode has its antinode in that position. The plate was excited with harmonic force at a predefined level with a frequency lower than the fundamental natural frequency. The excitation frequency was then increased in steps (0.05 Hz) while keeping the predefined force within the tolerance specified ($\pm 0.5\%$) using feedback controller. After this test was completed, the excitation frequency was decreased from a frequency higher than the fundamental natural frequency in a similar way. At the end of the experimental test, every force level has two curves: one obtained increasing and the other one decreasing the excitation frequency. These two curves should be superimposed in case of linear response of the system, but present jumps and hysteresis in case of nonlinear response. Every time that the excitation frequency is changed during the measurement, 40 periods of the signal are discharged in order to remove the transient response of the system. The next 40 periods are recorded and used to calculate the vibration amplitude and phase at that excitation frequency. The time domain signals are measured at the sampling frequency of 6400 Hz. The typical measurement setup is explained in Fig. 6(b).

2.3. Linear results

The set of FRFs measured from the plate was used to extract the modal parameters. The first four mode shapes are shown in Fig. 7. In the figure caption, the number n and m of half-waves in x and y direction, respectively, of the mode shape is given. The natural frequencies and damping ratios are reported in Table 1. The fundamental mode of the plate is at 17.1 Hz and its damping ratio is 2.98%. A Finite Element Model (FEM) of the plate was also created using shell elements in ANSYS®. The storage modulus of the material at 17.1 Hz is used as the Young's modulus (51 MPa) in the FE model. The results from the FEM are in good agreement with the experimental results, as shown Table 1.

2.4. Nonlinear results

The fundamental mode was chosen for the large-amplitude vibration study. Six force levels (0.1, 0.25, 0.5, 0.75, 1.5 and 2.5 N) were selected and the nonlinear frequency–response curves of the plate were obtained. The amplitudes of the experimental responses are plotted in the Fig. 8 and the corresponding phases are shown in Fig. 9. The excitation frequency ω is non-dimensionalized by the natural frequency of the fundamental mode Ω in both figures. Similarly, the vibration amplitude is non-dimensionalized by the plate thickness. The two lowest force levels (0.1 and 0.25 N) show a linear behaviour and the frequency of the peak is practically the same as the linear natural frequency. However, from the third force level onwards, the resonance frequency increases, which is typical of hardening-type nonlinear systems. The largest two force levels (1.5 and 2.5 N) display significant hysteresis and jumps in both the amplitude and phase curves. A maximum vibration amplitude

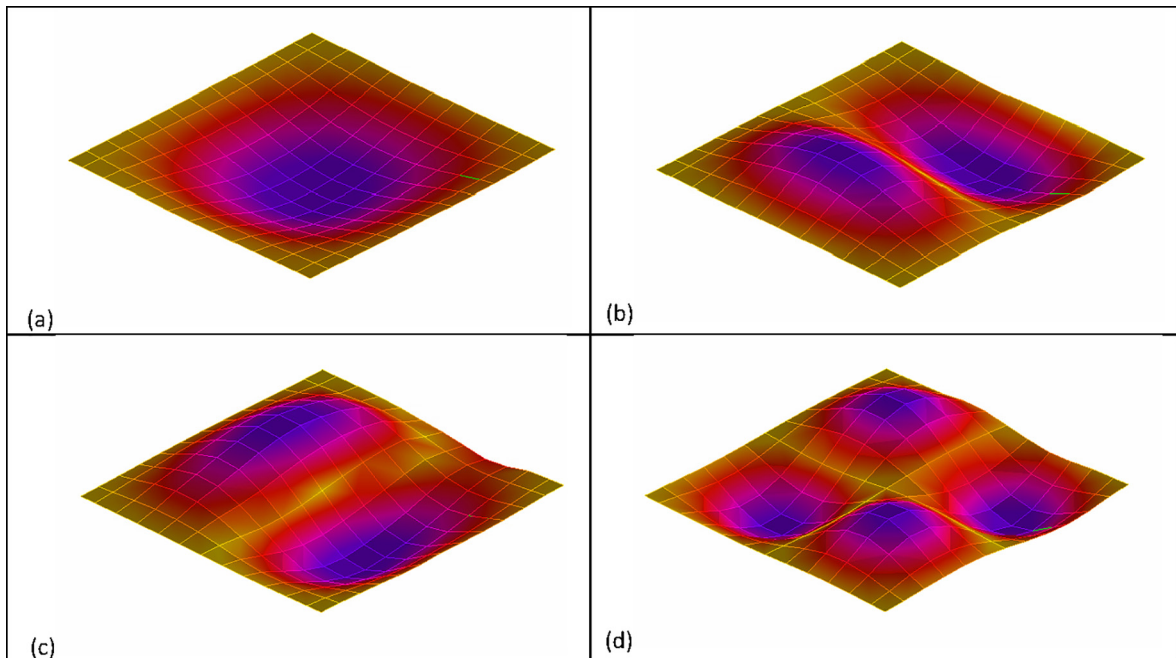
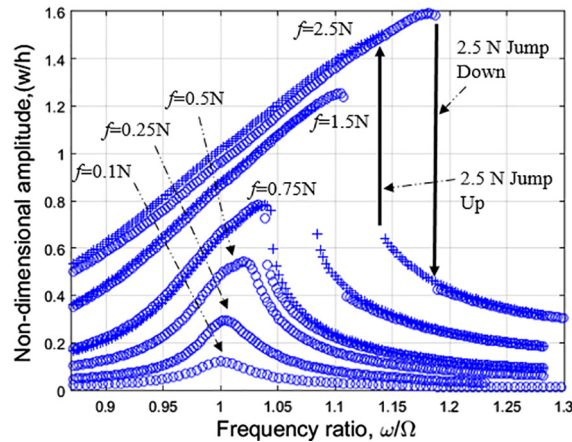


Fig. 7. Mode shapes of the glued rubber plate. (a) 1st mode ($n = 1$, $m = 1$); (b) 2nd mode ($n = 2$, $m = 1$); (c) 3rd mode ($n = 1$, $m = 2$); (d) 4th mode ($n = 2$, $m = 2$).

Table 1

Experimental modal analysis results for the rubber plate compared with numerical results.

Mode	Experimental Data		ANSYS	ROM (linear)
	Frequency, Hz	Damping Ratio, %	Frequency, Hz	Frequency, Hz
1	17.10	2.98	17.85	17.34
2	35.67	3.17	36.48	37.14
3	37.45	3.25	36.48	37.14
4	54.78	3.11	53.69	55.63

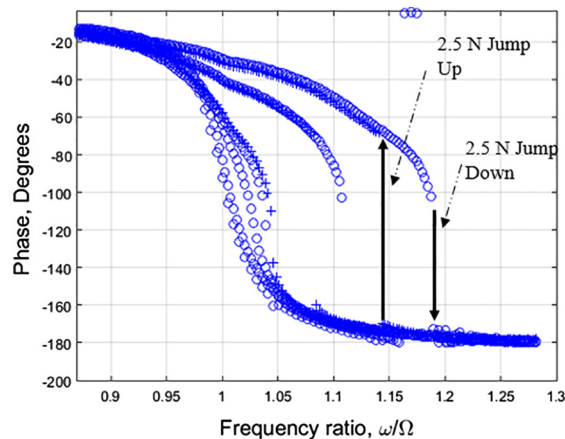
**Fig. 8.** Amplitude of the nonlinear response of the rubber plate. o, up curves (increasing excitation frequency); +, down curves (decreasing excitation frequency).

of 1.6 times the thickness was achieved for the highest force level (2.5 N). The increase in the resonance frequency is almost 20% from the natural frequency.

3. Reduced order model (MODEL 1)

3.1. Theory

A viscoelastic rectangular plate with coordinate system ($O; x; y; z$) having the origin (O) at one corner is considered, as presented in Fig. 10. The plate dimensions in x and y directions are a and b , respectively, and the thickness is h . The displacements of an arbitrary point of coordinates (x, y) on the middle surface of the plate are denoted by u, v and w in the x, y and z directions, respectively. Initial geometric imperfections in z direction of the plate are denoted by w_0 . Shear deformation and

**Fig. 9.** Phase of the nonlinear response of the rubber plate. o, up curves (increasing excitation frequency); +, down curves (decreasing excitation frequency).

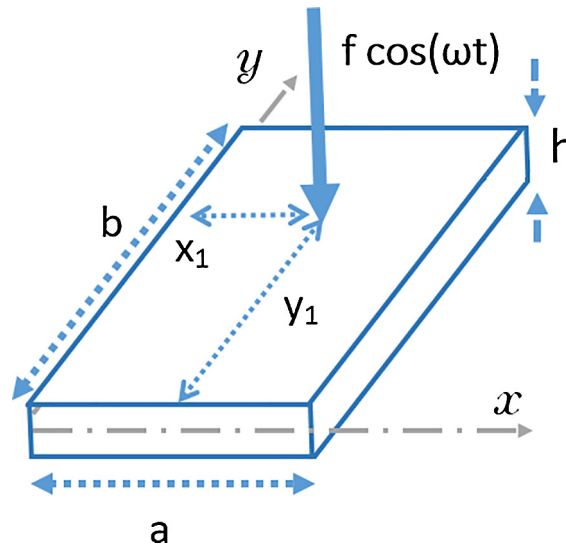


Fig. 10. Scheme of the rubber plate under study.

rotary inertia are neglected since the plate tested here is very thin ($h \ll a, b$) and their effects on the dynamics is negligible. According to the von Kármán strain–displacement relationship, the strain components $\varepsilon_x, \varepsilon_y, \gamma_{xy}$ at an arbitrary point of the plate are related to the middle surface strains $\varepsilon_{x,0}, \varepsilon_{y,0}, \gamma_{xy,0}$ and to the changes of curvature and torsion of the middle surface k_x, k_y, k_{xy} by

$$\varepsilon_x = \varepsilon_{x,0} + zk_x, \quad \varepsilon_y = \varepsilon_{y,0} + zk_y, \quad \gamma_{xy} = \gamma_{xy,0} + zk_{xy}, \quad (1)$$

$$\varepsilon_{x,0} = \frac{\partial u}{\partial x} + \frac{1}{2} \left(\frac{\partial w}{\partial x} \right)^2, \quad \varepsilon_{y,0} = \frac{\partial v}{\partial y} + \frac{1}{2} \left(\frac{\partial w}{\partial y} \right)^2, \quad \gamma_{xy,0} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial x} \frac{\partial w}{\partial y}, \quad (2)$$

$$\varepsilon_{x,0} = \frac{\partial u}{\partial x} + \frac{1}{2} \left(\frac{\partial w}{\partial x} \right)^2 + \frac{\partial w}{\partial x} \frac{\partial w_0}{\partial x}, \quad \varepsilon_{y,0} = \frac{\partial v}{\partial y} + \frac{1}{2} \left(\frac{\partial w}{\partial y} \right)^2 + \frac{\partial w}{\partial y} \frac{\partial w_0}{\partial y},$$

$$\gamma_{xy,0} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial x} \frac{\partial w}{\partial y} + \frac{\partial w}{\partial x} \frac{\partial w_0}{\partial y} + \frac{\partial w_0}{\partial x} \frac{\partial w}{\partial y}, \quad (3)$$

$$k_x = -\frac{\partial^2 w}{\partial x^2}, \quad k_y = -\frac{\partial^2 w}{\partial y^2}, \quad k_{xy} = -2 \frac{\partial^2 w}{\partial x \partial y}. \quad (4)$$

The kinetic energy of the plate is given by

$$T = \frac{1}{2} \rho h \int_0^a \int_0^b (\dot{u}^2 + \dot{v}^2 + \dot{w}^2) dx dy \quad (5)$$

where ρ is the density of the plate material (kg/m^3) and the over-dot indicates the time derivative. The stress–strain relationship is assumed to be following the linear elasticity constitutive equation for plane stress

$$\begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{bmatrix} = \begin{bmatrix} \frac{E}{1-\nu^2} & \frac{\nu E}{1-\nu^2} & 0 \\ \frac{\nu E}{1-\nu^2} & \frac{E}{1-\nu^2} & 0 \\ 0 & 0 & \frac{E}{2(1+\nu)} \end{bmatrix} \begin{bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{bmatrix}, \quad (6)$$

where E is Young's Modulus and ν is Poisson ratio. Since it is assumed that the transverse stresses are negligible, the elastic potential energy of the plate is given by

$$U_p = \frac{1}{2} \int_{-\frac{h}{2}}^{\frac{h}{2}} \int_0^a \int_0^b (\sigma_x \varepsilon_x + \sigma_y \varepsilon_y + \tau_{xy} \gamma_{xy}) dx dy dz. \quad (7)$$

Using the constitutive equation (6) and relationships (1), the potential energy of the plate can be rewritten as

$$U_p = \frac{Eh}{2(1-\nu^2)} \int_0^a \int_0^b \left(\varepsilon_{x,0}^2 + \varepsilon_{y,0}^2 + 2\nu\varepsilon_{x,0}\varepsilon_{y,0} + \frac{1-\nu}{2}\gamma_{xy,0}^2 \right) dx dy + \frac{Eh^3}{2(12(1-\nu^2))} \int_0^a \int_0^b \left(k_x^2 + k_y^2 + 2\nu k_x k_y + \frac{1-\nu}{2} k_{xy}^2 \right) dx dy + O(h^4). \quad (8)$$

The virtual work done by the external transversal harmonic force is given by

$$W = \int_0^a \int_0^b f \cos(\omega t) \delta(x - x_1) \delta(y - y_1) w dx dy = f \cos(\omega t) (w)_{x=x_1, y=y_1}, \quad (9)$$

where f is the amplitude of the harmonic point load, applied at the point $(x = x_1, y = y_1)$, and δ is Dirac delta function. The non-conservative damping forces are assumed of viscous type and are taken into account by using the Rayleigh's dissipation function given

$$F = \frac{1}{2} c \int_0^a \int_0^b (\dot{u}^2 + \dot{v}^2 + \dot{w}^2) dx dy, \quad (10)$$

where the viscous damping coefficient c is assumed to be M_y function of the mode shape, i.e. on the integer numbers n and m . In particular, $c_{m,n} = 2m_{m,n}\omega_{m,n}\zeta_{m,n}$, where $m_{m,n}$, $\omega_{m,n}$ and $\zeta_{m,n}$ are the mass, the natural frequency and the damping ratio of the corresponding mode (m, n) .

The boundary conditions of the considered plate are given by

$$w = N_x = N_{xy} = 0, \quad M_x = \pm k \frac{\partial w}{\partial x}, \quad \text{at } x = 0, a, \quad (11a-d)$$

$$w = N_y = N_{xy} = 0, \quad M_y = \pm k \frac{\partial w}{\partial y}, \quad \text{at } y = 0, b. \quad (11e-h)$$

The minus sign in Eqs. (11d,h) applies at the boundaries $x = a$ and $y = b$. Eqs. (11d,h) give any rotational constraint from zero bending moment ($M_x = 0$ and $= 0$, unconstrained rotation, obtained for $k = 0$) to zero rotation ($\partial w / \partial x = 0$ and $\partial w / \partial y = 0$, obtained as limit for $k \rightarrow \infty$), according to the value of k , where k is the stiffness per unit length of the elastic, distributed rotational springs placed at the four edges. In the present study, a very large value of k is assumed in order to approximate clamped boundary conditions for the out-of-plane displacement w , while the in-plane displacements u and v are assumed to be free at the edges.

In order to discretize the system, the mid surface displacements u , v and w are expanded by using trial functions, which satisfy the geometric boundary conditions (11a,e),

$$u(x, y, t) = \sum_{m=1}^M \sum_{n=1}^N u_{m,n}(t) \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right), \quad v(x, y, t) = \sum_{m=1}^M \sum_{n=1}^N v_{m,n}(t) \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right), \\ w(x, y, t) = \sum_{m=1}^M \sum_{n=1}^N w_{m,n}(t) \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right), \quad (11)$$

where $u_{m,n}(t)$, $v_{m,n}(t)$ and $w_{m,n}(t)$ are the generalized coordinates and t is the time. M and N the number of terms used to expand the corresponding displacements in the two directions. The initial geometric imperfections of the plate in z direction are assumed to be given by

$$w_0(x, y) = \sum_{m=1}^M \sum_{n=1}^N A_{m,n} \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right). \quad (12)$$

The elastic energy accumulated by the distributed rotational springs, which ensure the rotational constraint, can be written as

$$U_R = \frac{1}{2} \int_0^b k \left[\left(\left(\frac{\partial w}{\partial x} \right)_{x=0} \right)^2 + \left(\left(\frac{\partial w}{\partial x} \right)_{x=a} \right)^2 \right] dy + \frac{1}{2} \int_0^a k \left[\left(\left(\frac{\partial w}{\partial y} \right)_{y=0} \right)^2 + \left(\left(\frac{\partial w}{\partial y} \right)_{y=b} \right)^2 \right] dx. \quad (13)$$

The following notation is introduced for brevity,

$$\mathbf{q} = \{u_{1,1}, \dots, v_{1,1}, \dots, w_{1,1}, \dots, w_{m,n}\}^T. \quad (14)$$

The generic element of the time-dependent vector of the generalized coordinates $\mathbf{q}$ is referred as q_j ; the dimension of $\mathbf{q}$ is equal to the number N_q of degrees of freedom used in the mode expansion. The generalized forces Q_j are obtained by differentiating the virtual work done by external forces and the Rayleigh dissipation function

$$Q_j = \frac{\partial W}{\partial q_j} - \frac{\partial F}{\partial \dot{q}_j}, \quad \text{for } j = 1, \dots, N_q. \quad (15)$$

The Lagrange equations of motion are given by

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} + \frac{\partial U}{\partial q_j} = Q_j, \quad \text{for } j = 1, \dots, N_q, \quad (16)$$

where $U = U_p + U_R$. Eq. (16) can be rewritten in the following matrix form,

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{C}\dot{\mathbf{q}} + [\mathbf{K} + \mathbf{K}_2(\mathbf{q}) + \mathbf{K}_3(\mathbf{q}, \mathbf{q})]\mathbf{q} = \mathbf{f}_0 \cos(\omega t), \quad (17)$$

where $\mathbf{M}$ is the diagonal mass matrix of dimension $N_q \times N_q$; $\mathbf{C}$ is the linear viscous damping matrix, $\mathbf{K}$ is the linear stiffness matrix, $\mathbf{K}_2$ is the quadratic nonlinear stiffness terms, $\mathbf{K}_3$ denotes the cubic nonlinear stiffness terms and $\mathbf{f}_0$ is the vector representing the projection of the concentrated harmonic force on the generalized coordinates.

3.2. Numerical implementation

In order to obtain the equations of motion in a suitable form for numerical implementation, the system is multiplied by the inverse of the mass matrix and then is rewritten in the state-space form as follows,

$$\dot{\mathbf{q}} = \mathbf{y}, \dot{\mathbf{y}} = -\mathbf{M}^{-1}\mathbf{C}\mathbf{y} - \mathbf{M}^{-1}[\mathbf{K} + \mathbf{K}_2(\mathbf{q}) + \mathbf{K}_3(\mathbf{q}, \mathbf{q})]\mathbf{q} + \mathbf{M}^{-1}\mathbf{f}_0 \cos(\omega t). \quad (18)$$

The viscous damping matrix is assumed to be given by

$$\mathbf{M}^{-1}\mathbf{C} = \begin{bmatrix} 2\omega_{1,1}\zeta_{1,1} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 2\omega_{m,n}\zeta_{m,n} \end{bmatrix}, \quad (19)$$

where $\omega_{m,n}$ is the natural frequency of the mode (m, n) and $\zeta_{m,n}$ is the corresponding modal damping ratio. The software *Mathematica* [23] has been used to perform the surface integrals and to obtain the $2 \times N_q$ first-order ordinary differential equations (ODEs) written in state space form, as shown in Eq. (18). They are very long expressions containing nonlinear stiffness. Nondimensionalization of variables has been carried out for computational convenience; the vibration amplitude has been divided by the plate thickness and time has been multiplied by natural frequency. These $2 \times N_q$ first-order ODEs are solved by using the bifurcation and continuation software AUTO [24]. The AUTO software uses pseudo arc-length continuation and collocation methods to follow the solution path. It is important to note that, along with stable solutions, unstable solutions are also obtained. In particular, the analysis has been started by considering the excitation force as the bifurcation parameter, having fixed the excitation frequency far away from the resonance. The solution starts at zero amplitude of excitation and is increased slowly to reach the desired harmonic force level. Then, the solution is continued by varying the previously fixed excitation frequency, spanning the frequency range around the resonance in order to obtain the frequency–response curves.

3.3. Results and discussion

Initially, a linear analytical model with 507 degrees of freedom (DOFs) was developed. The storage modulus of the rubber is a function of frequency (typical of viscoelastic materials), as shown in Fig. 2. The storage modulus at the first natural frequency of the plate (17.1 Hz) was used as Young's modulus (51 MPa) in the model. The stiffness of the distributed rotational springs was fixed at 2000 N/rad, which was found by convergence analysis to accurately simulate fixed rotation. The natural frequencies of the linear analytical model are matching closely with the experimentally identified ones, as shown in Table 1.

In the nonlinear analysis, a reduced-order model with 39 DOFs, which are the number of terms inserted in the expansion of u, v given in Eq. (11), is used. In particular, the generalized coordinates are

$$W_{1,1}, W_{3,1}, W_{5,1}, W_{7,1}, W_{1,3}, W_{3,3}, W_{5,3}, W_{7,3}, W_{1,5}, W_{3,5}, W_{5,5}, W_{1,7}, W_{3,7}, u_{2,1}, u_{4,1}, u_{6,1}, u_{8,1}, u_{2,3}, u_{4,3}, u_{6,3}, u_{8,3}, u_{2,5}, u_{4,5}, u_{6,5}, u_{2,7}, u_{4,7}, v_{1,2}, v_{3,2}, v_{5,2}, v_{7,2}, v_{1,4}, v_{3,4}, v_{5,4}, v_{7,4}, v_{1,6}, v_{3,6}, v_{5,6}, v_{1,8}, v_{3,8}.$$

Table 2

Detailed modal expansion of the measured surface of the rubber plate.

Mode $(\bar{M}, \bar{N})$	(1,1)	(1,2)	(2,1)	(2,2)	(1,3)	(3,1)	(3,3)
$\frac{A_{\bar{M}, \bar{N}}}{h}$	0.45	0.07	−0.03	−0.02	0.18	0.07	0.03

The measured surface of rubber plate (see Fig. 5) was approximated using 7 fundamental modes of the plate and its respective coefficients are given in Table 2. The coefficient of the fundamental mode is $A_{1,1} = 0.45 h$, which is much higher when compared to other modes' coefficients. Hence, the geometric imperfection of the plate was introduced in the model with the fundamental mode only using Eq. (12) ($\tilde{M} = 1$; $\tilde{N} = 1$) [25]. The nonlinear vibration response was obtained by the computer program AUTO for all the six force levels (0.1, 0.25, 0.5, 0.75, 1.5 and 2.5 N). Lowest two force levels show typical linear response; the damping ratio used for these two levels matches with the experimental modal analysis results ($\zeta = 0.031$). After these two force levels, the response presents nonlinearities. At the third and fourth force levels ($f = 0.5$ and 0.75 N), the resonance frequency (i.e. the frequency of the peak of the response) is higher than the linear natural frequency confirming the typical hardening nonlinear behaviour of plates. The linear viscous damping value was adjusted to fit each experimental response. At the last two force levels ($f = 1.5$ and 2.5 N), the responses present significant hysteresis and jumps. The comparison between the model and experiments is shown in Fig. 11. They match well and the model captures the nonlinear response of the system accurately. It is interesting to point out that the geometric imperfection reduces the hardening behaviour of the plate [26]. The variation of the damping ratios used to matching the response at each force level is plotted against the maximum amplitude (obtained at the peak) of vibration reached at that specific excitation in Fig. 12. Specifically, the damping ratios are 0.031, 0.031, 0.033, 0.036, 0.042 and 0.048 for the six force levels in the respective order. The increase of the damping ratio with the vibration amplitude confirms earlier results [3,14,20]. The dissipation of the system is nonlinear and needs to be taken into account for better estimation of the nonlinear response. The increase of damping from its linear value is 60% when the vibration amplitude is 1.6 times the thickness of the plate.

4. Single degree of freedom approximation (MODEL 2 & 3)

4.1. Parameter identification

The geometrically nonlinear response of a structure around its fundamental mode can be approximated as single-degree-of-freedom (SDOF) system with nonlinear stiffness, i.e. a modified Duffing oscillator [27]. This assumption is valid for many cases, including plates and panels, when no internal resonances appear. Here, two SDOF models will be considered. Model 2: SDOF with classical linear viscous damping, shown in Fig. 13. Model 3: SDOF with nonlinear damping represented by the loss factor, shown in Fig. 14. Model 1 is reduced to Model 2 when the plate is approximated with a SDOF, as both uses linear viscous damping description.

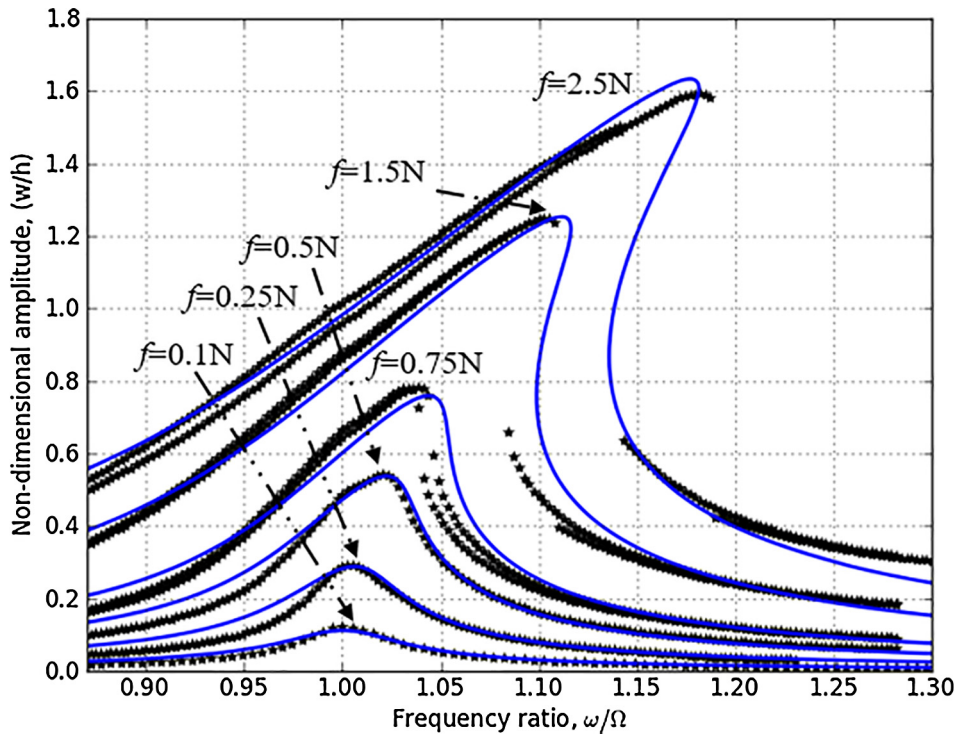


Fig. 11. Comparison between ROM (MODEL 1) and experimental results for the non-dimensional vibration amplitude versus the non-dimensional excitation frequency. —, ROM; *, experimental data.

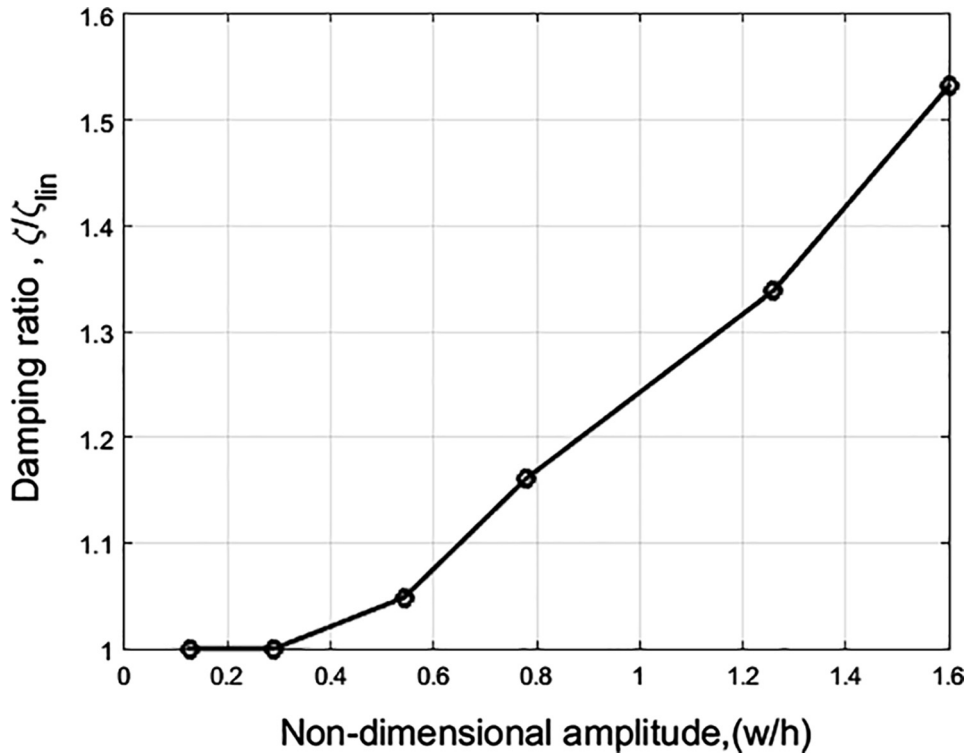


Fig. 12. Variation of the normalized viscous damping ratio (using MODEL 1) versus the vibration amplitude.

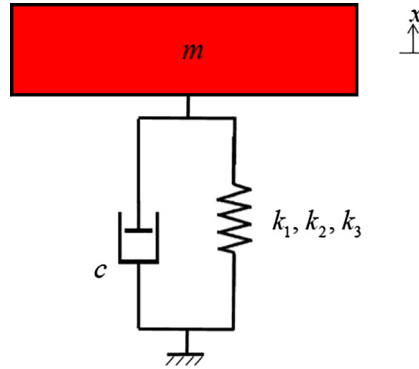


Fig. 13. Single degree of freedom model with linear viscous damping and geometrically nonlinear spring (MODEL 2).

4.2. SDOF with linear viscous damping (MODEL 2)

The equation of motion for the MODEL 2 is given by [8]

$$m\ddot{x}(t) + c\dot{x}(t) + k_1x(t) + k_2x(t)^2 + k_3x(t)^3 = pf(t), \quad (20)$$

where m is the modal mass, c is the viscous damping coefficient of the dashpot, k_1, k_2, k_3 are linear, quadric and cubic stiffness, respectively, $x(t)$ is displacement response, the over dot indicates differentiation with respect to time, $f(t)$ is the harmonic force and p is the modal projection coefficient (projection of the force on the mode). Various methods are available to fit experimental data to a dynamic model, and a comprehensive list can be found in [28]. Here the Temporal Method is chosen due to the availability of time domain data from experiments. As per the method, the response and applied force is expanded using truncated Fourier series with one harmonic component as below,

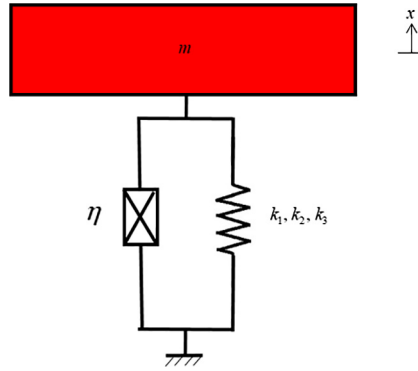


Fig. 14. Single degree of freedom model with dissipation represented by the loss factor (MODEL 3).

$$\begin{aligned} x(t) &= a_0 + a_1 \cos(\omega t) + a_2 \sin(\omega t), \\ f(t) &= b_0 + b_1 \cos(\omega t) + b_2 \sin(\omega t), \end{aligned} \quad (21)$$

where a_0 is the constant part of the displacement, a_1 and a_2 are cosine and sine coefficients of the expansion, b_0 is the constant part of the force, b_1 and b_2 are cosine and sine coefficients of the expansion of the excitation force. Additional harmonic terms can be considered to improve the accuracy of the model, but they do not give any improvement in the problem studied. In fact, the feedback controller used in the experiments acts only on the first harmonic, so that higher harmonics of the excitation can appear around the resonance. This means that, any higher harmonic detected in the response could be due to nonlinearity or generated by the distorted harmonic excitation. Therefore, the higher harmonics are discharged in the present identification since affected by measurement error.

Eq. (21) is non-dimensionalized using the following expressions

$$\tau = \omega t, \quad \tilde{x}(\tau) = x(t)/h, \quad (22)$$

which give

$$\begin{aligned} \tilde{x}(\tau) &= \tilde{a}_0 + \tilde{a}_1 \cos(\tau) + \tilde{a}_2 \sin(\tau), \\ f(\tau) &= b_0 + b_1 \cos(\tau) + b_2 \sin(\tau), \end{aligned} \quad (23)$$

where ω is the natural frequency in rad/s, t is the time in s, τ is the non-dimensional time, and the superimposed tilde indicates coefficients divided by the thickness h of the plate. Eq. (20) can be rewritten making use of (23), (24)

$$m\omega^2 h \ddot{\tilde{x}}(\tau) + c\omega h \dot{\tilde{x}}(\tau) + k_1 h \tilde{x}(\tau) + k_2 h^2 \tilde{x}(\tau)^2 + k_3 h^3 \tilde{x}(\tau)^3 = p f(\tau). \quad (24)$$

Dividing by $k_1 h$, the above equation can be rearranged as

$$\Omega^2 \ddot{\tilde{x}}(\tau) + 2\zeta \Omega \dot{\tilde{x}}(\tau) + \tilde{x}(\tau) + \beta_2 \tilde{x}(\tau)^2 + \beta_3 \tilde{x}(\tau)^3 = \lambda f(\tau), \quad (25)$$

where

$$\Omega = \frac{\omega}{\omega_n}, \quad \lambda = \frac{p}{m\omega_n^2 h}, \quad \omega_n = \sqrt{\frac{k_1}{m}}, \quad c = 2\zeta m\omega_n, \quad \beta_2 = \frac{k_2}{k_1} h, \quad \beta_3 = \frac{k_3}{k_1} h^2, \quad (26)$$

ω_n is the natural frequency in rad/s, ζ is the viscous damping ratio, Ω is the ratio between the excitation frequency and the natural frequency. To estimate the parameters λ and ζ , the non-dimensional nonlinear stiffness terms β_2 and β_3 are initially removed from Eq. (25), giving

$$2\zeta \Omega \dot{\tilde{x}}(\tau) - \lambda f(\tau) = -\Omega^2 \ddot{\tilde{x}}(\tau) - \tilde{x}(\tau). \quad (27)$$

Eq. (27) is valid for every time point. Hence, there are two unknowns and as many equations as the number n of data points for the linear experimental curves. The least square error minimization method can be employed to solve the following over constrained system and obtain ζ and λ that give the best fitting of all the linear time responses

$$\begin{bmatrix} 2\zeta \Omega \dot{\tilde{x}}(\tau_1) & -f(\tau_1) \\ 2\zeta \Omega \dot{\tilde{x}}(\tau_2) & -f(\tau_2) \\ \vdots & \vdots \\ 2\zeta \Omega \dot{\tilde{x}}(\tau_n) & -f(\tau_n) \end{bmatrix} \begin{bmatrix} \zeta \\ \lambda \end{bmatrix} = \begin{bmatrix} -\Omega^2 \ddot{\tilde{x}}(\tau_1) - \tilde{x}(\tau_1) \\ -\Omega^2 \ddot{\tilde{x}}(\tau_2) - \tilde{x}(\tau_2) \\ \vdots \\ -\Omega^2 \ddot{\tilde{x}}(\tau_n) - \tilde{x}(\tau_n) \end{bmatrix}. \quad (28)$$

Representing the above problem with the matrix $\mathbf{A}$ and two vectors $\mathbf{X}$ and $\mathbf{B}$, the error function can be created and minimized

$$err = \|\mathbf{A}\mathbf{X} - \mathbf{B}\|^2. \quad (29)$$

Since there are two force levels that can be considered linear in the experimental data set, the average λ from these two sets of data is used for further calculations. In the analysis of the nonlinear experimental curves, the nonlinear stiffness terms are retained and ζ , β_2 and β_3 are identified using the same procedure as above, while λ has been already identified. In particular,

$$2\zeta\Omega\dot{\tilde{x}}(\tau) + \beta_2\tilde{x}(\tau)^2 + \beta_3\tilde{x}(\tau)^3 = \lambda f(\tau) - \Omega^2\ddot{\tilde{x}}(\tau) - \tilde{x}(\tau), \quad (30)$$

which gives

$$\begin{bmatrix} 2\zeta\Omega\dot{\tilde{x}}(\tau_1) & \tilde{x}(\tau_1)^2 & \tilde{x}(\tau_1)^3 \\ 2\zeta\Omega\dot{\tilde{x}}(\tau_2) & \tilde{x}(\tau_2)^2 & \tilde{x}(\tau_2)^3 \\ \vdots & \vdots & \vdots \\ 2\zeta\Omega\dot{\tilde{x}}(\tau_n) & \tilde{x}(\tau_n)^2 & \tilde{x}(\tau_n)^3 \end{bmatrix} \begin{bmatrix} \zeta \\ \beta_2 \\ \beta_3 \end{bmatrix} = \begin{bmatrix} \lambda f(\tau_1) - \Omega^2\ddot{\tilde{x}}(\tau_1) - \tilde{x}(\tau_1) \\ \lambda f(\tau_2) - \Omega^2\ddot{\tilde{x}}(\tau_2) - \tilde{x}(\tau_2) \\ \vdots \\ \lambda f(\tau_n) - \Omega^2\ddot{\tilde{x}}(\tau_n) - \tilde{x}(\tau_n) \end{bmatrix}. \quad (31)$$

The parameter λ and the set of β_2 , β_3 and ζ , identified at each force level, fully characterize the plate nonlinear response in the frequency neighbourhood of the fundamental mode.

4.3. Loss factor with a SDOF (MODEL 3)

A nonlinear SDOF with nonlinear viscoelastic dissipation described by the loss factor (MODEL 3) is represented in Fig. 14. The viscoelastic (or hysteretic) damping is most commonly measured by the loss tangent and applied to linear systems, for which the loss tangent coincides with the loss factor. While the loss tangent is defined in linear regime, the loss factor is defined based on the ratio of dissipated and storage energies. This makes it valid for nonlinear systems such as biological and rubber structures, because the energies can still be calculated from the force–displacement loops [10,29]. The loss factor η , which is equal to the loss tangent $\tan \delta$ only for a linear system, is defined as

$$\eta = \frac{\Delta W_d}{2\pi W_s}, \quad (32)$$

where ΔW_d is the energy dissipated per cycle and W_s is the storage energy, also per cycle. It is useful to recall that the specific damping capacity is instead defined as $\Psi = \Delta W_d/W_s$.

The calculation of dissipated and storage energies is simple for linear systems. As shown in Fig. 15, both energies can be evaluated from the force and displacement data measured for a vibration cycle at any excitation frequency. The dissipated energy is represented by the area inside the force–displacement loop (thick line in Fig. 15). The shape of the loop is always an ellipse for linear systems. The storage energy is half of the area below the storage modulus line (shaded areas in Fig. 15, where the area for negative displacements is changed of sign to be positive). For linear systems, the areas W_{s1} and W_{s2} for traction and compression are the same. The storage energy is obtained as

$$W_s = \frac{W_{s1} + W_{s2}}{2}, \quad (33)$$

$$W_s = W_{s1} = W_{s2} = \frac{1}{2}k_1X^2, \quad (34)$$

where k_1 is the stiffness (or storage modulus) of the system, represented by the slope of the dashed line in Fig. 15, and X is the maximum displacement.

For nonlinear systems, the same procedure can be extended. The dissipated energy for nonlinear system is still represented by the area inside the force–displacement loop (thick line in Fig. 16). But, because of the nonlinearities in the response, this loop may not be an ellipse anymore and can exhibit complex shapes. The force–displacement loop for a hardening nonlinear system (like a plate) is shown in Fig. 16 at resonance. It is important to note that, as the frequency of excitation increases, the loop rotates clockwise in the force–displacement plane, for both linear and nonlinear systems. This is due to the phase change between force and displacement as the frequency of excitation changes. Nevertheless, the area can be calculated from the experimentally measured force and displacement if enough resolution in time is used.

The storage energy of the structure can also be estimated as shown in Fig. 16, by extending the definition for linear systems. For nonlinear systems, the areas for negative and positive displacements can be different, especially for softening type systems. Eq. (33) is still valid and the storage energy can be defined as half of the area below storage modulus curve (dashed curve in Fig. 16). Here, for MODEL 3, the system parameters identified by MODEL 2, except the damping ratio, are used as the elastic part is common for both models. In particular, the non-dimensional nonlinear stiffness β_2 and β_3 are utilized to

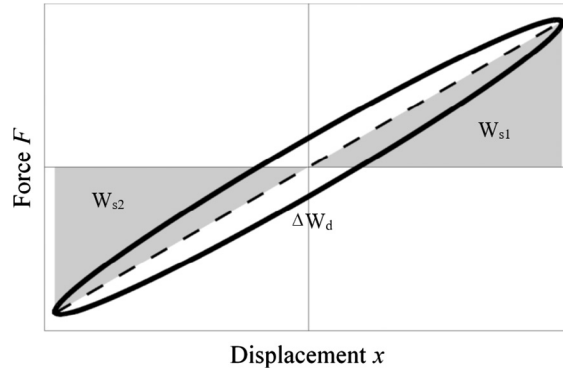


Fig. 15. Force-displacement loop for a linear system. The storage modulus is represented by the dashed line.

evaluate the non-dimensional storage energy (for convenience both storage and dissipated energy are evaluated using the non-dimensionalization introduced in Eq. (23))

$$W_{s1} = \frac{X_{\max}^2}{2} + \frac{\beta_2 X_{\max}^3}{3} + \frac{\beta_3 X_{\max}^4}{4}, W_{s2} = \frac{X_{\min}^2}{2} + \frac{\beta_2 X_{\min}^3}{3} + \frac{\beta_3 X_{\min}^4}{4}, \quad (35)$$

where $X_{\min}$ (negative in Fig. 16) and $X_{\max}$ are the non-dimensional minimum and maximum vibration amplitudes in a cycle, respectively. From the two energies, the loss factor of the system can be calculated using Eqs. (33), (34), which gives a measure of the nonlinear viscoelastic damping of the plate vibrating at the fundamental mode.

4.4. Parameter identification procedure

An identification tool was constructed using the programming language MATLAB based on the Temporal Method. This tool makes use of the time domain data measured during the nonlinear response measurement by stepped sine tests. Typical time domain data (voltage, force, displacement and frequency) measured from a stepped sine test at constant force excitation are shown in Fig. 17. It is interesting to observe that the voltage necessary to drive the electrodynamic exciter during a test varies significantly, even if the force is kept constant within $\pm 5\%$, by the feedback controller. Therefore, keeping the voltage constant during stepped sine tests is not a good practice since it introduces different excitation levels in the same experiment. The duration of one test could be anywhere from a few minutes to a few hours, depending upon the frequency range, frequency resolution and difficulty in ensuring the force level within the tolerance specified.

During the stepped sine test, the initial frequency of excitation is chosen away from the resonance and slowly increased (UP curve)/decreased (DOWN curve) in steps while ensuring constant force via closed loop feedback. This is necessary to capture the multiple stable solutions appearing around the resonance, which is a characteristic of nonlinear systems. The force and displacement time series are chopped into time segments pertaining to one particular frequency of excitation. The number of cycles at every frequency step may vary because of the feedback control, but there are at least 80 cycles (40 cycles for eliminating the transient vibration and 40 cycles to capture the steady state vibration). Hence the tool uses the last 40 cycles for estimating the energies of the plate.

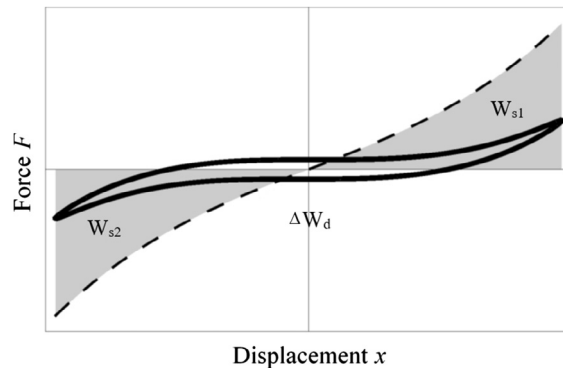


Fig. 16. Force-displacement loop for an hardening nonlinear system. The storage modulus is represented by the dashed curve.

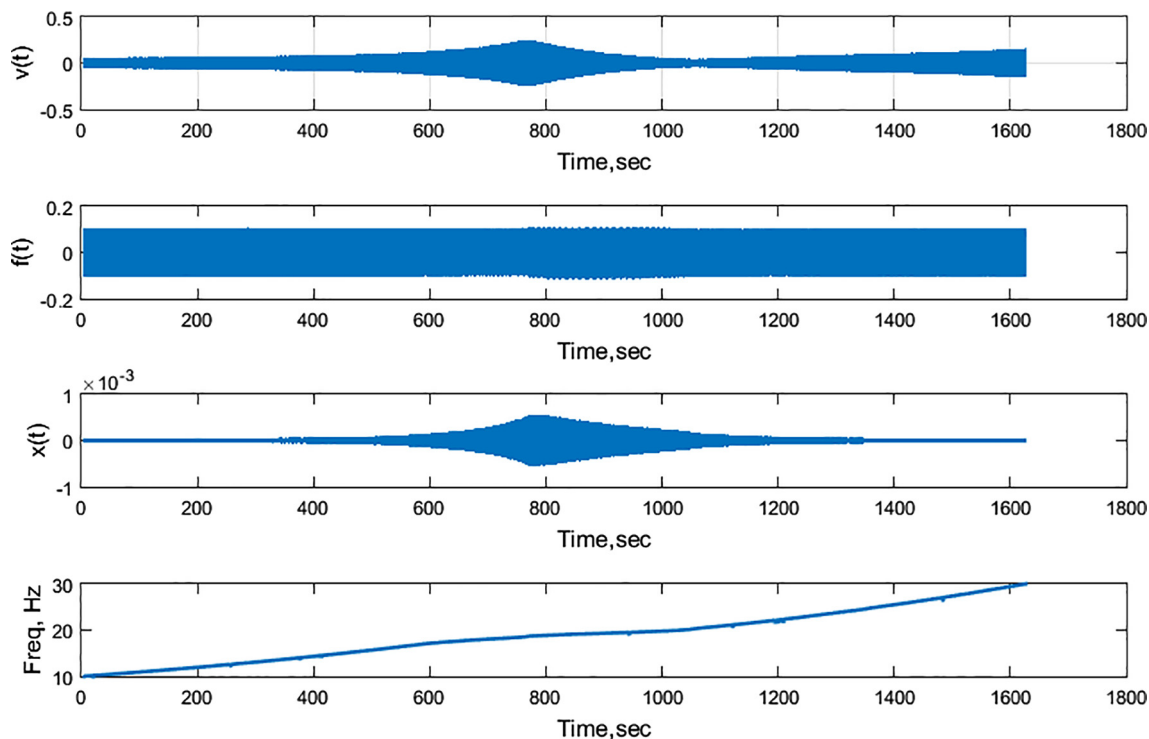


Fig. 17. Typical time domain data measured from a stepped sine vibration test. $V(t)$, voltage used to drive the shaker; $f(t)$, measured force excitation; $x(t)$, dynamic displacement at the center of the plate; Freq., excitation frequency.

Table 3
Identified system parameters by Model 2 and Model 4.

	ζ	β_2	β_3	$\hat{\beta}_2$	$\hat{\beta}_3$	Force, N
Model 2	0.0232	0	0.215	–	–	0.1
	0.0232					0.25
	0.0254					0.5
	0.0271					0.75
	0.032					1.5
	0.038					2.5
Model 4	0.0232	0	0.21	0	0.39	–

4.5. Results and discussion

4.5.1. SDOF with linear viscous damping (MODEL 2)

From the measured data, the parameters defining the system are obtained using the Temporal Method. The identified parameters are given in Table 3. These parameters are inserted in Eq. (25) and the response of the plate is calculated using integration methods and plotted against the respective experimental response in Fig. 18. A reasonably good agreement between the SDOF model's response and experimental data is found. The damping ratio ζ , identified using this model, is plotted against the maximum vibration amplitude and shown in Fig. 19. The trend and values are similar to those obtained by MODEL 1, thus reinforcing the validity of the SDOF identification tool. It is advantageous to have such a simple tool to extract the damping from experimental nonlinear response without building a complex and computationally expensive ROM as MODEL 1. However, the simple tool provided by MODEL 2 does not have such limitation and can be used to identify the damping in a nonlinear structure, as long as the nonlinear response involves one predominant mode.

4.5.2. Loss factor in a SDOF (MODEL 3)

The dissipated energy for the plate is shown in Fig. 20. The storage energy was calculated by using Eq. (35) and plotted in Fig. 21. The dissipated and storage energies increase as the structure undergoes large amplitude vibrations. The loss factor calculated as the ratio of the two energies is plotted in Fig. 22. The loss factor of the plate for the two lowest excitation levels is not constant and decreases with the excitation frequency. It gradually drops with a constant slope and equals to twice the

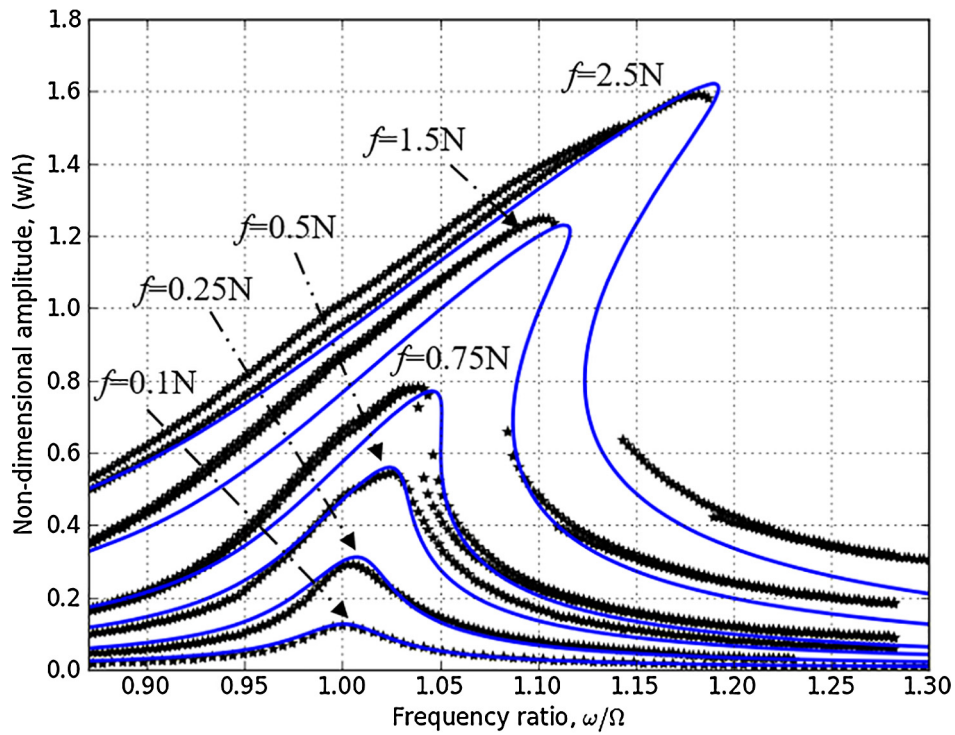


Fig. 18. Comparison between MODEL 2 and experimental results for the non-dimensional vibration amplitude versus the non-dimensional excitation frequency. —, MODEL 2; *, experimental data.

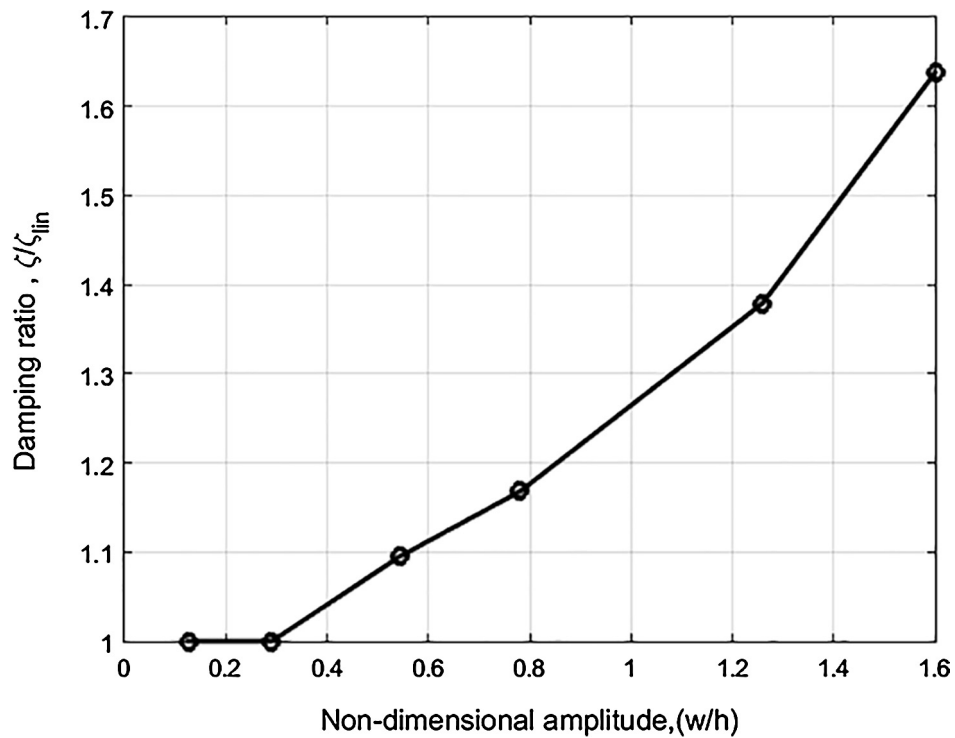


Fig. 19. Variation of the normalized viscous damping ratio (using MODEL 2) versus the vibration amplitude.

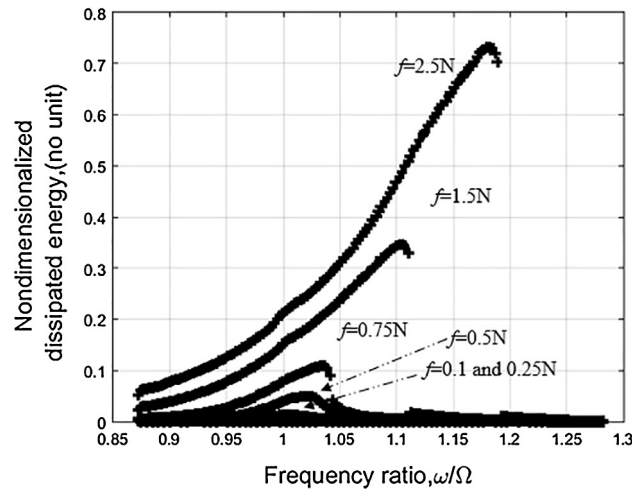


Fig. 20. Dissipated energy at the different harmonic force levels versus excitation frequency.

value of the previously identified damping ratio at the linear resonance, confirming the theory. As the plate undergoes non-linear vibrations (from 0.5 N onwards), the loss factor coincides with the linear curves away from resonance and significantly increases as it approaches the resonance. This confirms that the dissipation energy increases more than the storage energy in large amplitude vibrations. The identified loss factor at the resonance, for each force level, is 0.048, 0.048, 0.053, 0.055, 0.061 and 0.072, from the lowest to the highest force. The loss factor increases around 50% from its linear value.

MODEL 3 identifies the damping at every frequency and it is based on the energy ratio of the system; so, it can be used for systems which cannot be described by simple damping models. This has the potential to be used also in highly damped, biological and rubber structures where the material is also physically nonlinear (hyperelastic and viscoelastic). The loss factor cannot be directly used as a nonlinear damping model since it is not associated to an expression to be inserted in the equation of motion, differently from the complex modulus which is its linear counterpart.

5. Standard linear solid model (MODEL 4)

The last model used to identify nonlinear damping is derived from the standard solid model of linear viscoelasticity, where geometric nonlinearity is introduced in both the springs. The model is shown in Fig. 23. The details of the formulation are given in [30]. The stress–strain relationship for the standard linear solid model is

$$\sigma + \tau_r \frac{d\sigma}{dt} = E_2 \left(\varepsilon + \tau_r \frac{d\varepsilon}{dt} \right) + E_1 \tau_r \frac{d\varepsilon}{dt}, \quad (36)$$

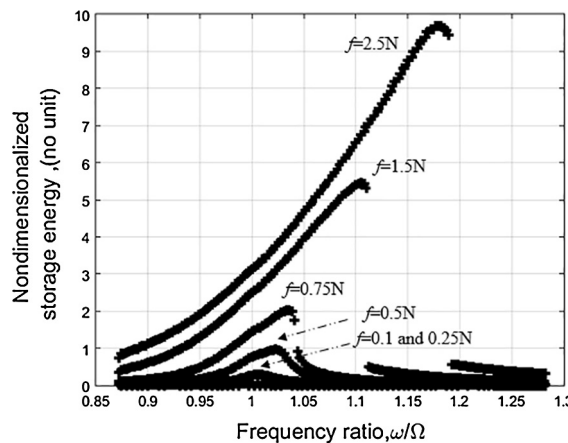


Fig. 21. Storage energy at the different harmonic force levels versus excitation frequency.

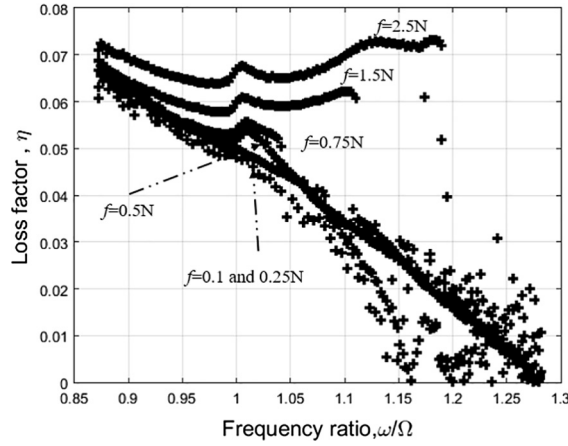


Fig. 22. Loss factor at the different harmonic force levels versus excitation frequency.

where σ is the stress, ε the strain, E_1 and E_2 the Young moduli of the two springs of the mechanically equivalent model in Fig. 23, $\tau_r = \eta/E_1$ is the viscoelastic relaxation time constant, and η is the viscosity parameter of the dashpot. The nonlinear damping introduced by this model is proportional to $x^2\dot{x}$, which is coherent with the expression introduced in reference [20]. Detailed explanation about the identification method used can be found in [21,30]. The SDOF model takes the form [30]

$$\ddot{x}(t) + \frac{2\zeta\omega_n}{1 + (2\zeta\omega/\omega_n)^2} \left[\dot{x}(t) + 2\frac{\tilde{\beta}_2}{h}x(t)\dot{x}(t) + 3\frac{\tilde{\beta}_3}{h^2}x^2(t)\dot{x}(t) \right] + \omega_n^2 \left[x(t) + \frac{\beta_2}{h}x^2(t) + \frac{\beta_3}{h^2}x^3(t) \right] + \frac{(2\zeta\omega)^2}{1 + (2\zeta\omega/\omega_n)^2} \left[[x(t) - \text{avg}(x(t))] + \frac{\tilde{\beta}_2}{h}[x^2(t) - \text{avg}(x^2(t))] + \frac{\tilde{\beta}_3}{h^2}[x^3(t) - \text{avg}(x^3(t))] \right] = \frac{p}{m}f \sin(\omega t), \quad (37)$$

where avg indicates the average value of the function in the vibration period and

$$\frac{k_1}{m} = \omega_n^2, \beta_2 = \frac{k_2 h}{k_1}, \beta_3 = \frac{k_3 h^2}{k_1}, \tau_r = \frac{2\zeta}{\omega_n}, \quad (38a-d)$$

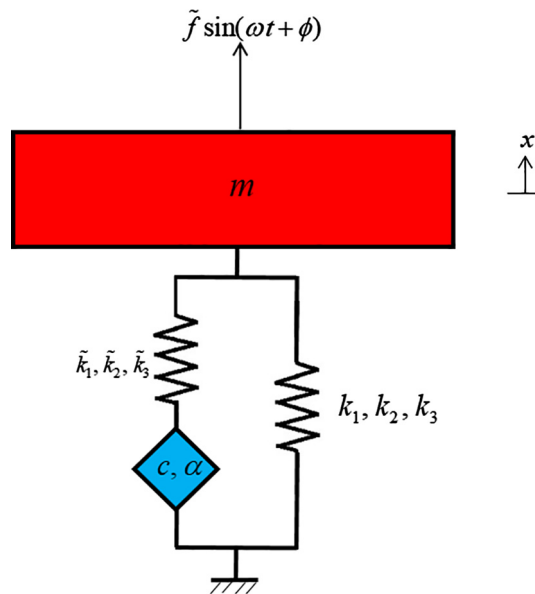


Fig. 23. SDOF fractional standard linear solid model with nonlinear springs (MODEL 4).

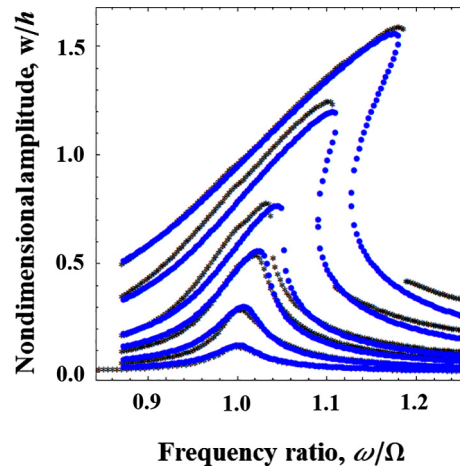


Fig. 24. Comparison between MODEL 4 and experimental results for the non-dimensional vibration amplitude versus the non-dimensional excitation frequency. ●, MODEL 4; *, experimental data.

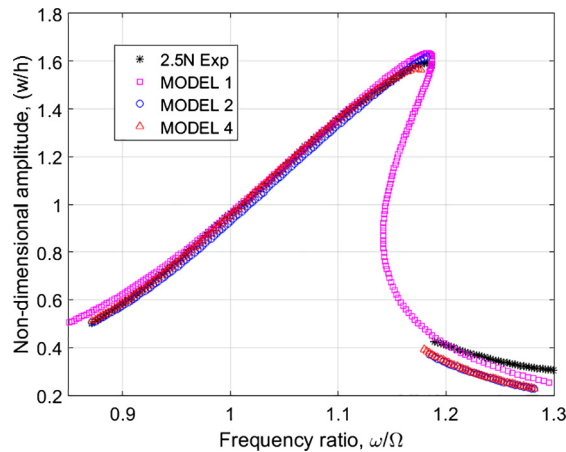


Fig. 25. Comparison between experimental and all three numerical models' responses at 2.5 N. *, experimental data; □ – MODEL 1; ○ – MODEL 2; △ – MODEL 4.

$$\tilde{k}_1 = k_1, \tilde{\beta}_2 = \frac{\tilde{k}_2 h}{k_1}, \tilde{\beta}_3 = \frac{\tilde{k}_3 h^2}{k_1}. \quad (38a-g)$$

$\beta_2, \beta_3, \tilde{\beta}_2, \tilde{\beta}_3$ are the non-dimensional nonlinear stiffness coefficients of the two geometrically nonlinear springs. It can be observed that the stiffness in Eq. (37) is a function of the vibration frequency ω , which also matches with the experimental results in Figs. 2 and 3 for the silicone material.

5.1. Results and discussion

The response from the model and experimental data are shown in Fig. 24. The close agreement between them confirms the accurate identification of nonlinear damping variation present in the experimental measurements. The identified parameters are given in Table 3. The quadratic and cubic stiffness terms are extremely similar between MODEL 2 and 4. The linear viscous damping of the MODEL 4 is identical to the experimental damping ratio. The single value description of damping covering the linear and nonlinear regime is particularly significant as it eliminates the necessity to adjust the damping depending upon the vibration amplitude.

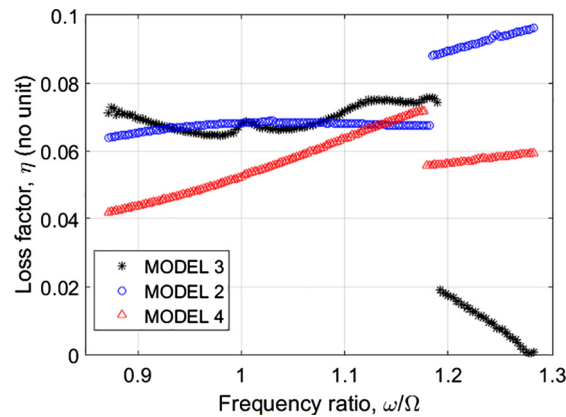


Fig. 26. Comparison between loss factors calculated using different models at 2.5 N. *, MODEL 3; o - MODEL 2; Δ - MODEL 4.

6. Comparison of all models

The comparison of the frequency responses obtained from the different models to the experimental data is shown in Fig. 25 for the highest force level reached in the tests (2.5 N). It can be seen that the three models identify the nonlinear stiffness of the rubber plate during large amplitude vibrations very well. In general, they are appropriate to capture the nonlinear stiffness exhibited by plates and curved panels during large amplitude vibrations. However, for MODEL 1 and MODEL 2, the damping ratio needs to be adjusted at each excitation level during large amplitude vibrations to correctly estimate the maximum vibration amplitude experienced by the structural element. This is not the case for the MODEL 4, as the damping model is inherently nonlinear and captures the damping variation shown by the rubber plate with good accuracy.

Experimentally identified loss factors are compared in Fig. 26 to the corresponding loss factors calculated from the numerical time responses of Model 2 and Model 4 for the 2.5 N excitation. The area inside the force–displacement loop was calculated from the time response by using Eqs. (25) and (37). It can be noticed that the numerically evaluated loss factors are similar to the experimental values only around the resonance peak of the plate. The loss factors are different between numerical simulations and experiments away from the resonance. This is acceptable in mostly of the applications since the influence of damping on the plate vibration response is significant only around the resonance. Away from resonance, mass and stiffness of the plate play a dominant role. However, the dissipation mechanics can be modelled by using a more sophisticated viscoelastic model to match the experimental values at all frequencies. This is left to a future study.

7. Conclusions

As the practical importance of understanding damping in large amplitude vibrations grows, several applicable damping descriptions were compared to the reference case of the damping identification based on a reduced order model with linear viscous damping. Experiments were performed on a square rubber plate made of silicone to measure its nonlinear response and consequently capture its nonlinear damping, which increases with the excitation level. State-of-the-art experimental techniques were used with success to record the nonlinear response of the structure, which features a hardening behaviour. A non-contact measurement of the plate profile was also performed to take into account geometric imperfections in the model. The reduced order model captures the nonlinear dynamics of the rubber plate very well. The damping values estimated by matching the peak amplitude of the experimental response at each force level show nonlinear variation, confirming earlier studies. The increase of damping is found to be around 60% when the vibration amplitude is 1.6 times the plate thickness. Three SDOF models were fit to the experimental data, each one with a different damping description. All the three procedures are capable to identify the linear and nonlinear response and damping of the plate. If viscous damping is considered, the damping value has to be increased with the forcing amplitude since the damping description cannot take into account the nonlinear variation. The loss factor does not provide a damping model per se, but the identified results clearly show and larger increase of the dissipated energy, with respect to the storage energy, in case of large amplitude vibrations. The standard linear solid model with geometrically nonlinear springs, originally introduced by Amabili [21,30], introduces a nonlinear damping model which is capable to reproduce numerical results in agreement with experiments for the full range of harmonic force excitation investigated without any need to adjust the dissipation parameters. It can be concluded that the MODEL 4 is the preferred single-degree-of-freedom tool to identify the nonlinear stiffness and damping of a panel during large amplitude vibrations.

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